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(54) **BIOLUBRIFICANTES FOR LARGE MEDIUM
AND SMALL INDUSTRIAL COMPRESSORS**

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ABSTRACT

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The invention is based on the development of biolubricants from vegetable oils, whose performance is superior to traditional lubricants currently used on the market, the vast majority of which are of petrochemical origin. Another great positive aspect is the rapid biodegradability, in less than 28 days more than 60% of the product biodegrades. In most countries, the disposal use after of these petrochemical products is prohibited, which creates a major problem and cost. The flexibility and the possibility of assembling molecules with different configurations generate lubricants with previously unimaginable performance, such as, for example, high lubricity, adjustable viscosity and low freezing points.

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BIOLUBRICANTES FOR LARGE MEDIUM AND SMALL INDUSTRIAL COMPRESSORS

FIELD OF THE INVENTION

[0001] The invention is based on the development of biolubricants from vegetable oils, whose performance is superior to traditional lubricants currently used on the market, the vast majority of which are of petrochemical origin. Another great positive aspect is its biodegradability and lower environmental impact.

[0002] In most countries, the disposal after use of these products of petrochemical origin is prohibited, which creates a major problem and cost.

[0003] The flexibility and possibility of assembling molecules with different configurations generate lubricants with previously unimaginable performance, such as, for example, lubricity, viscosity and freezing point.

BACKGROUND OF THE INVENTION

[0004] Lubricants play an important role in the lifetime of equipment whose main function is to reduce friction and wear, thereby maintaining integrity and reducing maintenance costs. Furthermore, when correctly applied, they minimize damage caused by high temperatures, corrosion, and friction. In the opposite condition, at low temperatures, they allow liquids to flow through small-thick pipes, preventing them from clogging, as is the case with refrigerator and air-conditioning compressors.

[0005] The main applications of lubricants are for industrial and domestic compressors (refrigerators and air conditioning), vehicle engines, wind farm engines, whose demand is increasing in volume and performance. The main function of reducing friction without causing an increase in energy is to mainly reduce environmental impacts.

[0006] Most industrial lubricants are of petrochemical origin based mainly on mineral oils with different degrees of viscosity.

[0007] Another class are synthetic lubricants based on polyol esters (pentaerythritol, neopentyl glycol, trimethylolpropane, polypropylene glycol) and polyalkylene glycols (PAG's), all from petrochemical origin. This class of products is of high quality and cost, but has low degradability, which makes the disposal of these materials difficult.

[0008] Another very relevant aspect are the labor processes arising in activities mainly where mineral oils that contain carcinogenic substances are handled due to the presence of polyaromatic substances.

DESCRIPTION OF THE INVENTION

[0009] Mainly aiming to reduce the environmental impacts of lubricants, 2 types of biolubricants were developed with viscosities of 68 centistokes and 46 centistokes for use in industrial compressors.

[0010] All formulations developed have a minimum content of 90% of vegetable origin with different attributes regarding friction reduction, lubricity, lifetime, thermal and oxidative resistance, and cost.

[0011] The precursor materials of vegetal origin for the products were palm kernel oil, palm oil, soybean oil mainly, whose participation by weight is above 90%.

[0012] The remainder of the composition whose share is less than 10% are polyols, such as pentaerythritol, neopentyl glycol, trimethylolpropane, propylene glycol.

[0013] In one embodiment, the present invention relates to a process for producing polyol ester formulations from palm kernel, palm, soybean, rapeseed, sunflower, corn, peanut, castor oil and their respective fatty acids or methyl esters or ethyl.

[0014] Vegetable or animal oils are in natural form or in the form of fatty acids, esters, hydrogenated, epoxidized, oxidized, dehydrogenated, or polymerized.

[0015] Fatty materials make up 70 to 80% by weight of the formulation, preferably above 90%.

[0016] The polyols used in the formulations were pentaerythritol, neopentyl glycol, trimethylolpropane, propylene glycol, di-propylene glycol, polypropylene glycol, ethylene glycol, diethylene glycol, polyethylene glycol, whose composition by weight is below 30%, preferably below 10%.

[0017] Polyol ester formulations synthesized by the process are also described and can be formulated alone or in mixtures to obtain the desired characteristics, both from a performance and economic point of view.

[0018] The synthesized polyol ester formulations can be used in large, medium, and small industrial compressors.

[0019] Polyester formulations are biodegradable and can be discarded after use or recovered due to their high oxidative stability.

EXAMPLE 1

- [0020] 1) Load 700 grams of palm kernel methyl ester (3.24 mol) and 300 grams of dimerized ester (0.98 mol) into the reactor;
- [0021] 2) Add 143 grams of pentaerythritol (1.04 mol);
- [0022] 3) Add 5.48 grams of catalyst;
- [0023] 4) Heat to 230° C. and react until approximately 90 grams of methanol distills;
- [0024] 5) Apply vacuum to react residual hydroxyls for 1 hour;
- [0025] 6) Cool to 90° C.;
- [0026] 7) Add 30 grams of clarifying earth, keep stirring for 30 minutes;
- [0027] 8) The viscosity at 40° C. measured on the Canon Fenske viscometer was 64.9 centistokes.

Physical-Chemical Analysis

Water	400 ppm
Acidity	0.45 mgkoh/gr
Viscosity at 40° C.	64.9 cst
Freezing point	-15° C.
Saponification index	183 mgkoh/g
Hydroxyl Index	13 mgkoh/g
Appearance	yellow liquid

EXAMPLE 2

- [0028] 1) Load 700 grams of distilled methyl dimerate (2.90 mol) into the reactor;
- [0029] 2) Add 301 grams of neopentyl glycol (2.90 mol);
- [0030] 3) Add 5.48 grams of catalyst;
- [0031] 4) Heat to 230° C. and react until approximately 90 grams of methanol distills;
- [0032] 5) Apply vacuum to react residual hydroxyls for 1 hour;

- [0033] 6) Cool to 90° C.
 [0034] 7) Add 30 grams of clarifying earth, keep stirring for 30 minutes;
 [0035] 8) The viscosity at 40° C. measured on the Canon Fenske viscometer was 40.9 centistokes.

Physical-Chemical Analysis

Water	300 ppm
Acidity	0.64 mgkoh/gr
Viscosity 40° C.	40.9 cst
Freezing point	-20° C.
Saponification index	175 mgkoh/g
Hydroxyl Index	11 mgkoh/g
Appearance	yellow liquid

EXAMPLE 3

- [0036] 1) Load 700 grams of isomerized methyl ester (2.28 mol) into the reactor;
 [0037] 2) Add 78 grams of pentaerythritol (0.57 mol);
 [0038] 3) Add 6.2 grams of catalyst;
 [0039] 4) Heat to 260° C. and react until approximately 70 grams of methanol distills;
 [0040] 5) Apply vacuum to react residual hydroxyls for 2 hours;
 [0041] 6) Cool to 90° C.;
 [0042] 7) Add 30 grams of clarifying earth, keep stirring for 30 minutes;
 [0043] 8) The viscosity at 40° C. measured on the Canon Fenske viscometer was 20.9 centistokes.

Physical-Chemical Analysis

Water	320 ppm
Acidity	0.24 mgkoh/g
Viscosity at 40° C.	20.9 cst
Freezing point	-30° C.
Saponification index	178 mgkoh/g
Hydroxyl Index	8 mgkoh/g
Appearance	colorless liquid

EXAMPLE 4

- 1) Load 700 grams of palm methyl ester (2.30 mol) into the reactor;
 [0044] 2) Add 102 grams of trimethylolpropane (0.77 mol);
 [0045] 3) Add 6.0 grams of catalyst;
 [0046] 4) Heat to 270° C. and react until approximately 70 grams of methanol distills;
 [0047] 5) Apply vacuum to react residual hydroxyls for 2 hours;
 [0048] 6) Cool to 90° C.;
 [0049] 7) Add 30 grams of clarifying earth, keep stirring for 30 minutes;
 [0050] 8) The viscosity at 40° C. measured on the Canon Fenske viscometer was 44.1 centistokes.

Physical-Chemical Analysis

Water	620 ppm
Acidity	0.74 mgkoh/g
Viscosity at 40° C.	44.1 cst
Freezing point	-10° C.
Saponification index	187 mgkoh/g
Hydroxyl Index	12.1 mgkoh/g
Appearance	yellowish liquid

EXAMPLE 5

- [0051] 1) Load 700 grams of dehydrated castor bean methyl ester (2.36 mol) into the reactor;
 [0052] 2) Add 85 grams of propylene glycol (1.12 mol);
 [0053] 3) Add 6.0 grams of catalyst;
 [0054] 4) Heat to 220° C. and react until approximately 68 grams of methanol distills;
 [0055] 5) Increase the temperature to 240° C., apply vacuum to react the residual hydroxyls for 2 hours;
 [0056] 6) Cool to 90° C.;
 [0057] 7) Add 30 grams of clarifying earth, keep stirring for 30 minutes;
 [0058] 8) The viscosity at 40° C. measured on the Canon Fenske viscometer was 56.2 centistokes.

Physical-Chemical Analysis

Water	320 ppm
Acidity	0.27 mgkoh/g
Viscosity at 40° C.	56.2 cst
Freezing point	-18° C.
Saponification index	192 mgkoh/g
Hydroxyl Index	10.1 mgkoh/g
Appearance	yellowish liquid.

1. Process for the production of polyol ester formulations characterized in that it is from palm kernel, palm, soybean, rapeseed, sunflower, corn, peanut, castor oil and their respective fatty acids or methyl or ethyl esters;

2. Process, according to claim 1, characterized in that the vegetable or animal oils are in natural form or in the form of fatty acids, esters, hydrogenated, epoxidized, oxidized, dehydrogenated or polymerized.

3. Process, according to claim 1, characterized in that the fatty materials make up 70 to 80% by weight of the formulation, preferably above 90%.

4. Process according to claim 1, characterized in that the polyols used in the formulations are pentaerythritol, neopentyl glycol, trimethylolpropane, propylene glycol, diethylene glycol, polypropylene glycol, ethylene glycol, diethylene glycol, polyethylene glycol, whose weight composition is below 30%, preferably below 10%.

5. Polyol ester formulations synthesized by the process as defined in claim 1, characterized in that they are formulated alone or in mixtures to obtain the desired characteristics, both from a performance and economic point of view.

6. Use of polyol ester formulations, as defined in claim 5, characterized in that they are for use in large, medium and small industrial compressors.

7. Polyester formulations, according to claim 5, characterized in that they are biodegradable and can be discarded after use or recovered due to their high oxidative stability.

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